

AEDLE: Designing Drama Therapy Interface for Improving Pragmatic Language Skills of Children with Autism Spectrum Disorder Using AR

Jungin Park*

School of Transdisciplinary Studies,
KAIST
Daejeon, Republic of Korea
pji0310@kaist.ac.kr

Gahyeon Bae*

Industrial Design, KAIST
Daejeon, Republic of Korea
bghblue@kaist.ac.kr

Jueon Park*

Industrial Design, KAIST
Daejeon, Republic of Korea
jueon0924@kaist.ac.kr

Seo Kyoung Park*

Industrial Design, KAIST
Daejeon, Republic of Korea
tjrud0330@kaist.ac.kr

Yeon Soo Kim*

Industrial Design, KAIST
Daejeon, Republic of Korea
ykim18@kaist.ac.kr

Sangsu Lee

Industrial Design, KAIST
Daejeon, Republic of Korea
sangsu.lee@kaist.ac.kr

ABSTRACT

This research proposes AEDLE, a new interface combining AR with drama therapy — an approved method of improving pragmatic language skills — to offer effective, universal, and accessible language therapy for children with Autism Spectrum Disorder (ASD). People with ASD commonly have a disability in pragmatic language and experience difficulty speaking. However, although therapy in childhood is necessary to prevent long-term social isolation due to such constraints, the limited number of therapists forbids doing so. Technology-based therapy can be a solution, but studies on utilizing digital therapy to improve pragmatic language are still insufficient. We conducted a preliminary user study with an ASD child and a therapist to investigate how the child with ASD reacts to drama therapy using AEDLE. We observed that our ASD child actively participated in AEDLE-mediated drama therapy, used our insights to recommend design suggestions for AR-based drama therapy, and explored various ways to utilize AEDLE.

CCS CONCEPTS

• **Human-centered computing** → **User interface design; Interface design prototyping; Accessibility systems and tools.**

KEYWORDS

Autism Spectrum Disorder, Pragmatic Language, Augmented Reality, Drama Therapy Tool

ACM Reference Format:

Jungin Park, Gahyeon Bae, Jueon Park, Seo Kyoung Park, Yeon Soo Kim, and Sangsu Lee. 2023. AEDLE: Designing Drama Therapy Interface for Improving Pragmatic Language Skills of Children with Autism Spectrum Disorder Using AR. In *Extended Abstracts of the 2023 CHI Conference on*

*These authors contributed equally to this research.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585809>

Human Factors in Computing Systems (CHI EA '23), April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585809>

1 INTRODUCTION

Therapy for improving pragmatic language skills is very important for children with Autism Spectrum Disorder (ASD) and must be performed early in childhood to minimize social isolation and prevent the long-term effects of pragmatic language difficulties [28]. ASD is a set of neurodevelopmental disorders characterized by disabilities in understanding the emotions of others, social interaction, communication, and a limited range of interests and behaviors [7, 19]. People with ASD have individual challenges and strengths in verbal language abilities but have a common difficulty in pragmatic language [32], which refers to the ability to use language in a social context [25]. It is an umbrella term for a variety of complex verbal and nonverbal abilities required in everyday conversations. These abilities include the ability to respond appropriately to speech, grasp the other person's intentions, maintain the conversation topic, or interact according to the situation context [19]. However, these are quite difficult for children with ASD (CWA) [19], causing CWA to feel more lonely and have difficulty forming friendships than typically developing peers [6, 24].

Many therapies have been developed to improve the pragmatic language skills of CWA, like pragmatic language interventions which focus on the aspects of pragmatics such as inference, exchange structure, turn-taking, and topic management [1]. However, although the role of an experienced therapist is crucial in conducting these interventions, these professionals are not always readily available due to their limited number and availability, thus many CWA face significant barriers in early and steady assistance. Therefore, the development of therapies that increase the effectiveness of school-based or home-based interventions is necessary [33]. Technology-based therapy can be a solution because it can deliver interventions at a low economic cost, and thus can become a universal and accessible therapy method independent from the availability of professional therapists [12]. Previous attempt to improve the pragmatic language skills of CWA under the age of 6 through a game via iPad application was studied in 2016 [12]. However, this

approach required more social development therapies to bridge the fundamental gap between the game and real-world interactions and thus was not as effective as anticipated [12].

Another emerging technology for consideration is Augmented Reality (AR), which is an interactive experience that combines computer-generated content and a real environment that allows computer graphic objects to appear in the surrounding environment [20]. Integrating AR in the field of technology-based therapy strategies can be a solution to mitigate the existing problems and consequently promote generalization — a phenomenon in which educated behavioral changes last for a long time in various situations and spread to related behaviors [22] — of educational content [21]. Therefore, there has been an attempt to utilize AR technology to support people with ASD to generalize the abilities they have learned in daily lives [35]. For example, a study in 2022 by Almurashi shows that therapy conducted using AR can promote continual enjoyment of therapy for the CWA [5]. However, there are only a limited number of studies that consider using AR specifically targeting to pragmatic language improvement for CWA.

This research focuses on developing an AR therapy environment to facilitate pragmatic language improvement for CWA by implementing drama therapy; one of the proven pragmatic language therapy methods. Drama therapy allows the participants to learn ways to manage and overcome real-life burdens by recalling the past experiences learned during therapy sessions [13]. According to past research, drama therapy also helps improve three pragmatic language abilities: introduction and responsiveness, non-verbal communication, and social-emotional attunement [4, 9]; which are three of five pragmatic language areas for children aged 5-11 [28]. Moreover, one important advantage of drama therapy is that it is conducted in small groups with children of the same age, helping CWA to become more entertained and motivated during learning [30]. In addition, since group therapy takes place in predictable theatrical situations, CWA who dislike unpredictable occasions can socialize and learn emotional and communal skills without feeling anxious [14]. Accordingly, the effectiveness of drama therapy in educating pragmatic language skills for CWA has been verified, and we expect an opportunity for this to be potentially developed within the AR environment. Since children are all easily immersed in imaginative play using stories [23] and drama therapy arouses children's interest for that matter, drama therapy can take advantage of using AR technology and creating a completely new space by adding computer graphics to the real world.

In short, we anticipate AR technology to offer CWA with a repetitive and effective educational environment for drama therapy at home or school without depending on the availability of professional therapists; thus mitigating the current problems of limited accessibility in early involvement of pragmatic language development. In this study, we propose a novel idea of an AR-based drama therapy interface called AEDLE (Autism Education Interface for Drama-therapy to Learn Expression) to help CWA improve their pragmatic language skills. We conducted a preliminary user study with one child with ASD using a Virtual Reality(VR)-based video prototype that provides an AR-like experience to investigate the response of the child. This preliminary user study was for investigating the feasibility and potential of AEDLE before building the

higher fidelity AR version to be deployed in a full study. We analyzed the responses of the ASD child and a post-interview from the professional therapist (who observed the child during the preliminary user study) to find suggestions for effective drama therapy interface design using AR technology and explore various ways to utilize AEDLE.

2 AEDLE DESIGN

2.1 Drama Therapy through AEDLE

AEDLE — short for Autism Education Interface for Drama-therapy to Learn Expression — is an AR-based drama therapy interface, and is pronounced [Ah-ee-deul], which means ‘children’ in Korean. AEDLE is developed with the intention to improve the pragmatic language skills of CWA by performing AR-driven drama therapy. Initially, our prototype of AEDLE is designed for CWAs aged 8-17 years but the story design is anticipated to be interchangeable according to the age group of the CWA user. It is designed to enable repeated learning through drama therapy at home or school, relatively independent from the availability of the therapist, allowing CWA's early intervention on pragmatic language skills. There are two areas of pragmatic language that can be improved through AEDLE, which are the ability to adjust and adapt according to the situation and non-verbal communication ability [27]. AEDLE is intended to be used by CWA groups (who can receive group therapy) with assistance from their parents or teachers who have a prior understanding of ASD behavior patterns; assistance is required mainly for safety in emergent situations. The following is the process of conducting group drama therapy through AEDLE, referring to commonly conducted drama therapy procedures. Before the therapy, a group of CWA learns stories used for the therapy. After sufficient learning of stories, the group of CWA and the assisting adult put on the AR Glass and proceed with the play following the instructions displayed on AEDLE. CWA highly vary in their ASD behavior patterns, and different children have their requirements for individualized learning. Here, professional therapists can assist with altering the level of instructions for each child to suit their behavioral patterns, enabling the personalized experience in group drama therapy. The most important aspect of therapy for CWA is the generalization process [17], which enables the application of learned skills to daily life. AEDLE uses a systematic fading technique [18] that sequentially removes instructions given during drama therapy for step-by-step generalization. The type and degree of fading program for each CWA can be decided by an experienced therapist at periodic check-ups. In conclusion, AEDLE is a device that enhances the accessibility of drama therapy by enabling non-professional parents or teachers to conduct therapies at schools and homes and allowing the personalization of group drama therapy experience for CWA to efficiently facilitate the generalization of what they have learned.

2.2 Interface Design of AEDLE

AEDLE is designed to make drama therapy more comfortable and enjoyable for CWA. To make our design comfortable for CWA who are sensitive to visual stimuli, we used simple shapes and avoided heavily sensory-loaded colors such as yellow [15]. The previous

case of graphic design decisions in games for people with intellectual disabilities including ASD describe clear communication of information to the user to be paramount [34]. Therefore, text in our interface have a large font size for legibility. In addition, facial expressions and conversation logs for verbalization (written text of conversations between characters in drama therapy) and behavioral instructions that CWA should follow are clearly defined with colors that distinguish each of them from other elements on the screen. Our design guides CWA to easily differentiate between the instructions directed to themselves from those that are directed to others. Various animation effects are repeatedly provided for intuitive learning and to stimulate continuous interest from the CWA.

Among the elements of pragmatic language learned in drama therapy, AEDLE particularly focuses on understanding conversations through reading written conversational logs to speak on their turn, understanding facial expressions through repeating the animated expressions accordingly and acknowledging expressions of other participants, and understanding behaviors by encouraging their reiteration.

The look of written conversation log (Fig. 1a) is designed to make play conditions to feel comfortable by making the flow of play predictable, considering that CWA prefer structured and predictable play conditions [8]. These instructions for verbalization are placed in common positions known in the Modified Reality (MR) design [29]. When it is the other child's turn, a speech box appears near the other child's left chest, and vocalized words appear in the speech box, animated word by word according to the speech speed. After that, the speech box will continue to exist until the other child has finished making facial expressions and following behaviors to signal the end of the turn.

The facial expression instruction (Fig. 1b) is designed to guide CWA to learn the relationship between facial expression and emotion and help make accurate facial expressions according to context. In a previous study on educational games for recognition of facial expressions, CWA preferred cartoon characters in comparison to characters that resembled real pictures [11]. So in our design, the participating children's faces are overlaid with AR character masks of the given role and can show accurate facial expressions. For CWA, it is more effective to train emotion recognition by looking at the features of partial facial components rather than the whole face [30]. Therefore in the facial expression instructions of AEDLE, a repetitive animation that emphasizes the facial components based on the Facial Action Coding System [10] are used to help CWA's facial expression recognition. The facial expression instructions are described with corresponding emotions; for example, the instruction "Smile because you are happy." is shown, expression and emotion words are emphasized with different colors to easily allow CWA to recognize the correlation between facial expressions and emotions. These instructions are consistently placed in commonly used positions known to be effective in the MR design [29].

Lastly, behavioral instruction in AEDLE (Fig. 1c) is designed to encourage CWA to naturally express their emotions and follow behaviors to empathize with others. We actively use animated effects such as moving graphics coming out from the characters for CWA to acknowledge and reflect to the given instruction at a glance. The behavioral instructions illustrate emotions and behaviors. Similar

to facial expression instructions, words representing behaviors and emotions are emphasized with color for easier recognition for the CWA. In addition, we use onomatopoeia and mimetic words that express behaviors that were familiar to children. The behavioral instructions are also consistently placed effective positions known in MR design [29].

3 METHOD

In this study, we conducted a preliminary user study to investigate if 1) AEDLE allows CWA's active participation in drama therapy without any sense of refusal and 2) AEDLE is sufficient for CWA to proceed with drama therapy with given instructions on screen.

3.1 Participants

The participants of the study were one child with ASD (Participant Child, herein PC) born in 2007 who have regular experience in group therapy. Additionally, our preliminary user study was conducted with one therapist (Participant Therapist, herein PT) who is experienced in conducting group drama therapy with CWA for three reasons: 1) to ensure the safety of PC during the preliminary user study, 2) to professionally interpret his behavior, and 3) to obtain an expert opinion on the AEDLE prototype. Research participants in this study were recruited with the cooperation of a local child development clinic in the Republic of Korea. PC has ASD grade 2 with an Intelligence Quotient (IQ) of 71 or less and a Global Assessment Scale (GAS) score of 21 to 40. PC can spontaneously speak words in short sentence levels and read and write sentences. PC has early communication skills (requesting, answering, following directions). At the time of the preliminary user study, PC was participating in role-play sessions to promote social interaction and improve appropriate communication skills (questioning, starting a dialogue, clarification) by recognizing the role in the group role-play.

3.2 Study Setting

3.2.1 Production of AEDLE Prototype. As a first step in exploring the potential of an AR-integrated drama therapy interface using AEDLE, a preliminary user study using a VR-based video prototype was used in this study. Before generating the video prototype, we investigated the interest of PC, such as favorite characters, elements, and colors through prior in-person and written interviews with PT one month before the experiment. Additionally, we sought advice from PT for each written instruction and story content. We finally created the VR prototype reflecting the information and advice gathered beforehand carefully in line with PC's ASD behavior pattern. The VR video prototype, which is PC's field of view, was filmed in the user study location using Samsung Gear 360 [31] to administer a similar experience as the AEDLE implemented in AR. Lastly, the 360-degree video was edited using Adobe Premiere Pro [3] and After Effect [2] to be viewed through the Oculus quest 2 [26] - a Head-Mounted-Display (HMD) VR device.

3.2.2 Preparation of the Preliminary Study. The location of the preliminary user study was provided by a local child development clinic which PC has been attending for ASD therapy for a long period of time. For PC to feel like he is undergoing real drama-therapy during our user study, visual scope in the video prototype

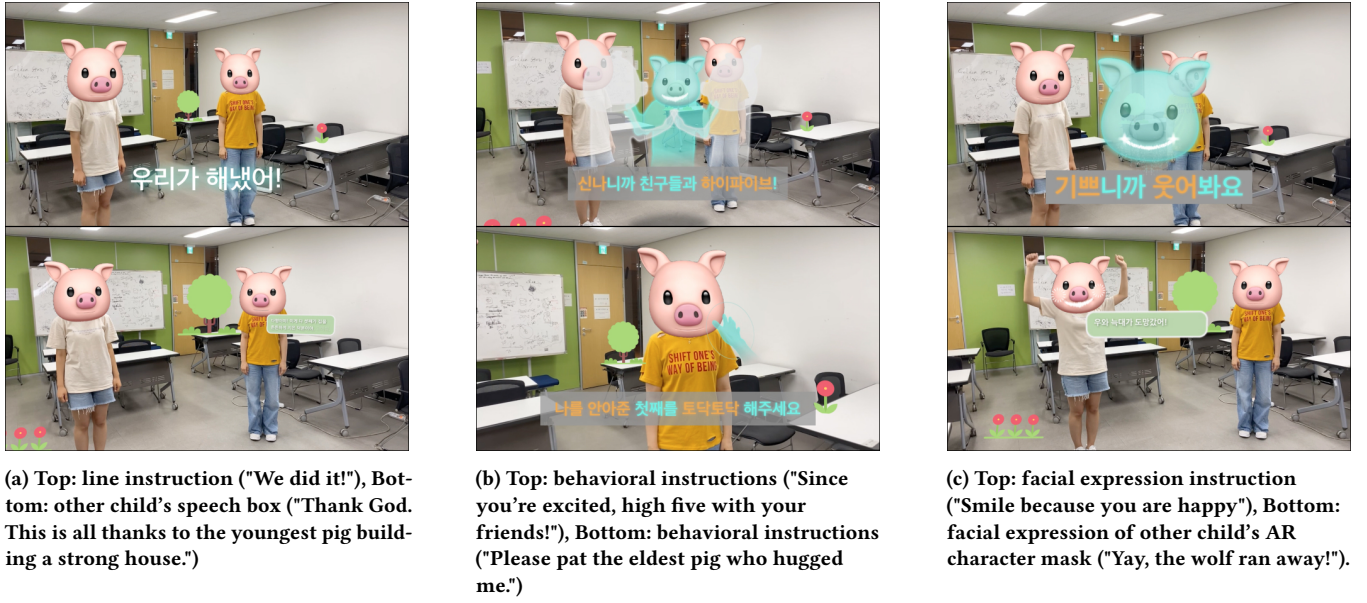


Figure 1: Examples of AEDLE design

is filmed closely mimicking the actual perspective of PC in the room. Subsequently, the preliminary user study was conducted at the place where we shot our video prototype, and the researchers who appeared in the VR video prototype wore the same clothes and were positioned in the same position as the pre-filmed video.

3.3 Procedure

First, the researchers gave prior explanation of the preliminary study to PT, then PT watched the VR video prototype wearing Oculus in advance to familiarize with the content of AEDLE. While PT listened to researchers' explanations, PC became familiar with the drama therapy stories by watching prepared videos. Before proceeding with the study, the surrounding environment was organized to ensure the safety of PC, and researchers who appeared in the video placed themselves in the same position as the pre-filmed VR video prototype. Then, PC was encouraged to spend time adapting to the Oculus device then put it on to conduct the study. During the user study, PC underwent drama therapy experience using our VR video prototype of AEDLE, and it was repeated three times with adequate rest in between. We tested our prototype three times to more reliably validate the applicability of AEDLE by allowing proper adaptation to the HMD device and unfamiliar therapy procedure for the child. PT helped PC wear the Oculus device and carefully observed the behavior of PC during the study. After the testing of the prototype, PT carried on a brief interview on PC with a series of questions prepared by the researchers. We requested PT, who was already familiar with PC, to carry out this post-interview on the child to make the interview session more comfortable and easy to follow. Subsequently, researchers interviewed PT about the applicability of drama therapy by using AEDLE. The study process and post-interviews were recorded with the consent of PC, PC's parent, and PT (Fig. 2a), and the screen view of PC was also recorded using

the Oculus mirroring function (Fig. 2b) to observe PC's behavior during the study. All preliminary user study procedures were designed and conducted with expert advice and performed with IRB approval.

4 FINDINGS AND DISCUSSION

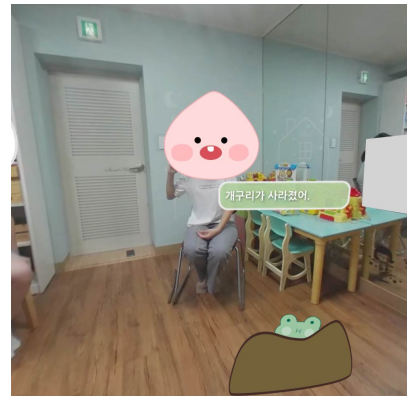
Qualitative data from the behavioral observation of PC and the interview results from both PC and PT were analyzed using the Affinity Diagram analysis method [16].

4.1 The Visual Experience of AR in Drama Therapy Can Stimulate the Interest of Children with Autism Spectrum Disorder.

On the whole, PC showed an interest in the drama therapy performed with AEDLE. This is particularly noteworthy regarding the user's preference for favorite characters, which led to a deeper interest in an AR-like environment with such characters. By analyzing the screen recording from Oculus, PC showed an interest in drama therapy content, and his perspective followed the animation elements as they appeared on the screen. For example, our participant's gaze followed the frog jumping from the floor and he fixed his attention on watching the animation of the facial expression instructions alongside the character masks. Likewise, we were able to observe PC's interest in AEDLE by observing PC talking about the characters he saw in the video and what the character said. Additionally, PC showed interest in the AR-like environment that he could experience in AEDLE by wearing the Oculus, turning his head from side to side in the seat, and turning back to look around the site of video prototype of VR graphics. During the post-interview with PT, when asked about the effectiveness of the AEDLE prototype, PT replied, *"(the therapy) was not bad because the child (PC) expressed interest,"* and agreed that PC participated with enthusiasm in the



(a) PC is wearing an Oculus and participating in the user study



(b) The screen that PC is viewing through the Oculus (The other child's speech box says, "The frog is gone.")

Figure 2: User study progression

drama therapy performed with AEDLE's prototype. When interviewing PC, to ensure PC understood and answered the questions, PT repeatedly asked questions using variety of language variations to ask regarding PC's interest, such as the following: "Did you enjoy today's program?" and "Do you think it would be fun to use this (Oculus) for drama therapy?". We observed that PC consistently answered positively, saying "It was fun." and "Yes (it was enjoyable)." In particular, PC's favorite characters used in the prototype were associated with his interest in this process.

4.2 Distinct and Diverse Instructions Can Make Children with ASD Understand Instructions and Engage in the Behavior.

We observed that PC recognized his turn and understood what the directed actions were but did not perform them. To address this, we suggest that different instruction designs should be attempted, including the use of conspicuous instructions, minimizing gaze movement, using more diverse and exaggerated visual stimuli, and providing additional sensory stimuli. The AEDLE user interface has several features, such as the other characters' speech boxes, their movements, and instructions PC has to follow - which PC moved his eyes and saw. PT said, "I think he understood. I think he knows he has to talk or act in his turn", suggesting that PC is aware of the meaning of given instructions as well as the existence of various elements within the AEDLE interface. However, not all of these resulted in action of the participant. For example, in the plot of the drama therapy, there was a scene where the other characters could not find the frog character as it was hiding in the pit. Therefore, PC had to follow the behavioral instructions which pointed at where the frog character was. PC looked at the other characters and shifted his gaze to the instruction when the behavioral instruction interface appeared. Thus, in this movement of the gaze, we can see that PC recognized his turn and the content of the instruction. However, PC did not follow the instructions on screen and only moved his eyes to the frog character, which was out of our anticipation in carrying out the behavioral instruction.

In conclusion, in our study, PC shows a low fulfillment rate compared to a high instruction recognition rate. Directions to solve to such phenomenon have been discussed based on videos showing PC's behavior, PT's observation records, and interviews with PT. First, we can try to use more conspicuous instructions. PT commented on the interviews and observational records, saying, "If the instructions look more significantly distinguishable from those of other characters, PC will not be able to miss them." Second, drama therapy using AEDLE is expected to continue by putting all the instructions and elements in the field of view of the characters. PT said, "the frog is seen on one side like this (pointing at the corner), so I think PC missed his line and could not follow it while looking at the frog". PT notes that several elements of the drama therapy content were not included in the field of view. Third, we can try using various exaggerated visual stimuli depending on the purpose. PT said the action in the behavioral instruction interface was not big enough, therefore it is not apparent what action the interface directs. Finally, it is necessary to address the fact that the current interface of AEDLE is focused only on visual stimuli. PT said, "Because PC has to recognize his turn by seeing only the visual interface, he cannot perform if he misses his line instructions. Therefore, I think it will be more difficult than common drama therapy." Similar to common drama therapy, it is necessary to consider providing additional stimulation of other senses, such as auditory stimulation.

4.3 AEDLE Has Potential as an AR-based Drama Therapy Tool.

From the observation of and interview with PT, it was possible to discover the applicability of AEDLE and gain insight into the application of drama therapy content and the generalization process to offer personalized therapy. First, AEDLE is expected to help repetitive therapy improve the pragmatic language skills of CWA in places that lack therapists, such as in homes. In particular, it can be helpful for CWA who lack pragmatic language skills, like talking in their turn or listening to others.

“If children practice a verbal response to a specific situation repeatedly at home or in therapy, some children memorize it almost entirely and express it well in same situations. So, I think it could be helpful for them.”

“I think it will be helpful for children to learn skills such as waiting when someone is talking, and then talking when they are done.”

If the content of the drama therapy using AEDLE is related to the situation that CWA likely experience in their daily lives, CWA can be trained to cope with real-life situations after the generalization process.

“When a mother is sick and says, ‘I feel quite exhausted now.’ or ‘I’m so sick.’ the child says ‘Don’t be sick.’, ‘Mom, I’m sad.’ and cries. If children perceive the relationship between situations and emotions, they will find out that people feel sad when someone is sick. Therefore, if the content relates to their daily lives like so, I think [CWA] will learn more when they do in drama therapy.”

Finally, it was discovered that drama therapy using AEDLE could provide personalized services to CWA. As each CWA has various ASD behavior patterns and characteristics, it would be beneficial that AEDLE offers adaptable difficulty adjustment by selecting instructions that will fade out for the generalization process.

“For those who are good at memorizing lines, remove the lines (instruction) first, and for those who are good at expressing emotions, remove behavioral elements (instruction) first. I hope it can be controlled separately. ... Each child has different characteristics, so I think it would be much better given that way.”

5 LIMITATIONS AND FUTURE WORK

This study uses a VR device as a research tool to apply the design of AEDLE to a preliminary user study with CWA. While watching the VR video prototype, PC frequently touched the part of his nose that was in contact with the VR device and said to the researchers that his nose was being pressed and uncomfortable. Therefore, the VR device used in this research, which is relatively heavy and inconvenient to wear compared to AR glasses, may have affected the study. In the future, a follow-up study using AR glasses is needed to confirm the usability of AEDLE. Moreover, there is a limitation in that the study was conducted using the video prototype. Finally, this study has a limitation stemming from its small sample size, as it was conducted with only one child with ASD and a therapist. As this study attempted a new concept of drama therapy using AR technology, it required primary confirmation of the possibility and gather initial insights and advice. Our long-term vision is to develop AEDLE that can interact with CWA using commercial AR devices and check if this allows for repeated drama therapy with parents or family members at home. To develop the AEDLE design further in future studies, we found a need to address the limitations in this study and verify the results.

First, more objective verification is needed to confirm whether the AEDLE design works as digital therapeutics for drama therapy with CWA. We plan to conduct user studies on a larger number of CWA and set quantitative indicators to analyze the data to confirm the results of their participation and the improvement in pragmatic language. In addition, group therapy in an AR environment will be conducted to observe the mutual interactions between actual CWA. We will check whether AEDLE helps CWA improve pragmatic language skills and generalization compared to existing drama therapies when the findings of this study are applied. Further, AR-based drama therapy has scalability. The users that can benefit from AEDLE for improving pragmatic language skills can be further diversified. Extending the age range of people with ASD and applicability to attention-deficit hyperactivity disorder (ADHD), borderline intellectual functioning, and intellectual disability, which require pragmatic language skills, can be considered. In addition, it is expected that the barriers to the accessibility of therapy and the needs of therapists can be reduced. Head-mounted display will be able to increase diverse interactions between HMD and CWA by recognizing the size and tone of the CWA’s voices and providing corresponding feedback according to their behavior; further follow-up studies are needed.

6 CONCLUSION

In this study, we proposed a new concept of AR-based drama therapy to improve the pragmatic language skills of CWA and created AEDLE, an AR-based drama therapy interface. To investigate the feasibility of this novel idea, we explored how the real end-user would react to our concept through conducting a preliminary user study with one child with ASD and a professional therapist. We analyzed the child’s behavior using a video prototype of AEDLE and interviewed the therapist for design suggestions and determined the potential of AEDLE. As a result, we discovered the positive applicability of AEDLE in real-life occasions to mitigate current burdens existing in early access of drama therapy to improve pragmatic language skills for children with ASD. Our CWA participant was interested in drama therapy along with animation elements in AEDLE; in particular, including characters CWA likes is considered as an important factor for initiating the child’s interest. Moreover, the visual instructions provided on AEDLE’s screen can support CWA to understand the context of drama therapy. The benefits of using AEDLE to perform drama therapy in real life can be summarised as the following; it can assist with the repetitive performance of interventions in the absence of professional therapists, which is necessary to improve pragmatic language in the long term, and drama therapy using AEDLE can be personalized and tailored to each CWA by adjusting the instruction or level of difficulty using pre-designed story sets. Additionally, we also discovered various possibilities of AEDLE in improving the effectiveness of drama therapy, such as easily allowing the generalization process by developing and designing a structured gradual fade out system of the instructions. Meanwhile, we derived suggestions for improving the design of instruction and elements, including visual cues and auditory stimuli, to actively trigger the behavior of CWA, by analyzing the result of our preliminary study. We expect that this preliminary study will serve as a starting point for follow-up studies that aim to

use technology to develop a system considering the improvement of pragmatic language of CWA and enhancing the effectiveness of home or school-based interventions.

ACKNOWLEDGMENTS

The authors would like to thank several people who have supported and helped us on our research. First, we thank Ji Hyung Seo, Chief of Iwaha Child Development Institute, for giving advice on experimental design and helping with our recruitment for the user study. Second, we thank Somin Choe, Therapist of Iwaha Child Development Institute, for helping to facilitate the user study procedures. Lastly, we thank Professor Kukju Kweon for giving professional feedback on our early design of AEDLE.

REFERENCES

- [1] Catherine Adams. 2003. Intervention for developmental pragmatic language impairments. *Aula Abierta* 82 (2003), 79–96.
- [2] Adobe. 2023. *Adobe After Effects*. Adobe. Retrieved March 5, 2023 from <https://www.adobe.com/kr/>
- [3] Adobe. 2023. *professional video editing software | adobe premiere pro*. Adobe. Retrieved March 5, 2023 from <https://www.adobe.com/products/premiere.html>
- [4] Catherine Aldred, Jonathan Green, and Catherine Adams. 2004. A new social communication intervention for children with autism: pilot randomised controlled treatment study suggesting effectiveness. *Journal of Child Psychology and Psychiatry* 45, 8 (2004), 1420–1430.
- [5] Haneen Almurashi, Rahma Bouaziz, Wallaa Alharthi, Mohammed Al-Sarem, Mohammed Hadwan, and Slim Kammoun. 2022. Augmented reality, serious games and picture exchange communication system for people with ASD: systematic literature review and future directions. *Sensors* 22, 3 (2022), 1250.
- [6] Nirit Bauminger and Connie Kasari. 2000. Loneliness and friendship in high-functioning children with autism. *Child development* 71, 2 (2000), 447–456.
- [7] Marcia Jean Carter. 2014. Diagnostic and statistical manual of mental disorders. *Therapeutic recreation journal* 48, 3 (2014), 275.
- [8] Monica Chester, Amanda L Richdale, and Jane McGillivray. 2019. Group-based social skills training with play for children on the autism spectrum. *Journal of Autism and Developmental Disorders* 49, 6 (2019), 2231–2242.
- [9] Blythe A Corbett, Alexandra P Key, Lydia Qualls, Stephanie Fecteau, Cassandra Newsom, Catherine Coke, and Paul Yoder. 2016. Improvement in social competence using a randomized trial of a theatre intervention for children with autism spectrum disorder. *Journal of autism and developmental disorders* 46, 2 (2016), 658–672.
- [10] P. Ekman and W.V. Friesen. 1978. *Facial action coding system: A technique for the measurement of facial movement*. Consulting Psychologists Press, Palo Alto, CA.
- [11] Tiago Fernandes, Samanta Alves, José Miranda, Cristina Queirós, and Verónica Orvalho. 2011. LIFEisGAME: A facial character animation system to help recognize facial expressions. In *ENTERprise Information Systems: International Conference, CENTERIS 2011, Vilamoura, Portugal, October 5-7, 2011, Proceedings, Part III*. Springer, International Conference on Enterprise Information Systems, Beijing, China, 423–432.
- [12] Sue Fletcher-Watson, Alexandra Petrou, Juliet Scott-Barrett, Pamela Dicks, Catherine Graham, Anne O'Hare, Helen Pain, and Helen McConachie. 2016. A trial of an iPad™ intervention targeting social communication skills in children with autism. *Autism* 20, 7 (2016), 771–782.
- [13] Loretta Gallo-Lopez and Lawrence C Rubin. 2012. *Play-based interventions for children and adolescents with autism spectrum disorders*. Routledge, .
- [14] Emma Godfrey and Deborah Haythorne. 2013. Benefits of dramatherapy for Autism Spectrum Disorder: a qualitative analysis of feedback from parents and teachers of clients attending Roundabout dramatherapy sessions in schools. *Dramatherapy* 35, 1 (2013), 20–28.
- [15] Marine Grandgeorge and Nobuo Masataka. 2016. Atypical color preference in children with autism spectrum disorder. *Frontiers in psychology* 7 (2016), 1976.
- [16] Bruce Hanington and Bella Martin. 2019. *Universal methods of design expanded and revised: 125 Ways to research complex problems, develop innovative ideas, and design effective solutions*. Rockport publishers, Beverly, Massachusetts.
- [17] Lee Seung Hee. 2022. A Model of the Relationship between Maintenance and Generalization in Applied Behavior Analysis. *Journal of Emotional & Behavioral Disorders* 38, 2 (2022), 45–62.
- [18] Lim Hee-Jung and Hong Joon-Pyo. 2019. Effect of Errorless learning versus Systematic fading in the mand and tact training on the speech acquisition for the children with autistic disorder. *Journal of Behavior Analysis and Support* 6, 1 (2019), 21–40.
- [19] BAE HEE-SOOK and Sun Hyung Lee. 2015. Group Language Therapy for Development of Pragmatic Skills of Children with ASD –DvT Theatrical Model Approach-. *The Journal of Drama(DR)* ., 45 (2015), 113–140. <https://doi.org/10.15716/dr.2015.45.113>
- [20] Mehmet Kesim and Yasin Ozarslan. 2012. Augmented reality in education: current technologies and the potential for education. *Procedia-social and behavioral sciences* 47 (2012), 297–302.
- [21] Julie A Kientz, Gillian R Hayes, Matthew S Goodwin, Mirko Gelsomini, and Gregory D Abowd. 2019. Interactive technologies and autism. *Synthesis lectures on assistive, rehabilitative, and health-preserving technologies* 9, 1 (2019), i–229.
- [22] Sunghee Lee. 2022. A Model of the Relationship between Maintenance and Generalization in Applied Behavior Analysis. *JOURNAL OF EMOTIONAL & BEHAVIORAL DISORDERS* 38, 2 (2022), 45–62. <https://doi.org/10.33770/JEBD.38.2.2>
- [23] Jeannie Lewis and Sube Banerjee. 2013. An investigation of the therapeutic potential of stories in Dramatherapy with young people with autistic spectrum disorder. *Dramatherapy* 35, 1 (2013), 29–42.
- [24] Jill Locke, Eric H Ishijima, Connie Kasari, and Nancy London. 2010. Loneliness, friendship quality and the social networks of adolescents with high-functioning autism in an inclusive school setting. *Journal of Research in Special Educational Needs* 10, 2 (2010), 74–81.
- [25] Danielle Matthews, Hannah Biney, and Kirsten Abbot-Smith. 2018. Individual differences in children's pragmatic ability: A review of associations with formal language, social cognition, and executive functions. *Language Learning and Development* 14, 3 (2018), 186–223.
- [26] Meta. 2023. *Oculus quest store: VR Games, Apps, & More*. Meta. Retrieved March 5, 2023 from <https://www.oculus.com>
- [27] S Oh, EJ Lee, and YT Kim. 2012. Preliminary study on developing test items of children's pragmatic language checklist. *Journal of Speech & Hearing Disorders* 21, 2 (2012), 111–135.
- [28] Lauren Parsons, Reinie Cordier, Natalie Munro, Annette Joosten, and Renee Speyer. 2017. A systematic review of pragmatic language interventions for children with autism spectrum disorder. *PLoS one* 12, 4 (2017), e0172242.
- [29] Robert Poláček. 2020. *User Interface Concept for Smart Glasses*. Ph. D. Dissertation. University of West Bohemia. <https://doi.org/10.13140/RG.2.2.26591.69288/1>
- [30] Christian Ryan and Cairiona Ní Charragáin. 2010. Teaching emotion recognition skills to children with autism. *Journal of autism and developmental disorders* 40, 12 (2010), 1505–1511.
- [31] Samsung. 2023. *Samsung Gear Series: 360*. Samsung. Retrieved March 5, 2023 from <https://www.samsung.com/sec/gear/360/>
- [32] Sarah Spencer. 2013. *Language Disorders from Infancy Through Adolescence: Listening, Speaking, Reading, Writing and Communicating* 4th edn Edited by RheaPaul and Courtenay F. Norbury (St. Louis, MO: Elsevier Mosby, 2012)[Pp. 756] ISBN 978-0-323-07184-0.£ 75.99.
- [33] Marcus L Thomeer, Rachael A Smith, Christopher Lopata, Martin A Volker, Alanna M Lipinski, Jonathan D Rodgers, Christin A McDonald, and Gloria K Lee. 2015. Randomized controlled trial of mind reading and in vivo rehearsal for high-functioning children with ASD. *Journal of autism and developmental disorders* 45, 7 (2015), 2115–2127.
- [34] Stavros Tsikinas and Stelios Xinogalos. 2018. Designing effective serious games for people with intellectual disabilities. In *2018 IEEE global engineering education conference (EDUCON)*. IEEE, 2018 IEEE Global Engineering Education Conference, Santa Cruz de Tenerife, Canary Islands, Spain, 1896–1903.
- [35] Francesco Vona, Mattia Gianotti, Alberto Patti, Jessica Barbieri, Lukasz Moskwa, Francesco Riccardo Di Gioia, and Franca Garzotto. 2022. Combining Virtual and Augmented Reality to improve daily autonomy for people with Autism Spectrum Disorder. In *Proceedings of the 2022 International Conference on Advanced Visual Interfaces*. Association for Computing Machinery, New York, NY, United States, 1–3.